IF A DIRECT SUM COMPUTES THE DERIVED FUNCTOR, SO DO THE SUMMANDS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5263-5288, following the statement at 5256-5261.

Editorial note. The omitted minor detail is the splitting of the displayed bijection along the direct sum decompositions of both sides; mechanical bookkeeping.

Editorial note. The final 'Similarly for LF' and 'Similarly for Y' follow the established mirror counting practice.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 5284 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Cofinality of pair resolutions. proof of lemma-direct-sum-defined (carded L26163, cross-referenced at ~5268) and Categories, Lemma categories-lemma-cofinal-essentially-constant. The pair index system (X/S) x (Y/S) is cofinal in (X+Y)/S.
- Essentially constant characterization. Categories, Lemma categories-lemma-characterize-essentially-constant-ind (cross-chapter, cited at ~5271 and ~5286). Characterization (4) converts the Hom bijectivity into definedness of RF at X.

Source reference. Original reference 05PY: focused-seeds/geometry/S9-der-direct-sum-computes/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A1S: focused-seeds/geometry/S9-der-direct-sum-computes/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SD: sources/stacks-derived/pinned-original-source.tex, source lines 5019–5019. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 05S8: sources/stacks-derived/pinned-original-source.tex, source lines 4736–4736. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Assumptions and notation as in Situation 05S8. Let X, Y be objects of $\mathcal{D}$. If $X \oplus Y$ computes RF, then X and Y compute RF. Similarly for LF.*

Proof. If $X \oplus Y$ computes RF, then $RF(X \oplus Y) = F(X) \oplus F(Y)$. In the proof of Lemma 05SD we have seen that the functor $X/S \times Y/S \to (X \oplus Y)/S$, $(s, s') \mapsto s \oplus s'$ is cofinal. Thus by Categories, Lemma 0A1S and by characterization (4) of Categories, Lemma 05PY we know that for any object W in $\mathcal{D}'$ the map

$$\mathrm{Hom}_{\mathcal{D}'}(F(X \oplus Y), W) \longrightarrow \mathrm{colim}_{s:X\to X', s':Y\to Y'} \mathrm{Hom}_{\mathcal{D}'}(F(X' \oplus Y'), W)$$

is bijective. Since this arrow is clearly compatible with direct sum decompositions on both sides, we conclude that the map

$$\mathrm{Hom}_{\mathcal{D}'}(F(X), W) \longrightarrow \mathrm{colim}_{s:X\to X'} \mathrm{Hom}_{\mathcal{D}'}(F(X'), W)$$

is bijective (minor detail omitted). Thus by Categories, Lemma 05PY we conclude RF is defined at X with value $F(X)$. Similarly for Y. □